METRIC

A-A-54580
20 April 1992

COMMERCIAL ITEM DESCRIPTION

FILES, PULP CANAL, DENTAL (RHOMBOIDAL)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-F-42001(DM) which is cancelled.

This commercial item description covers standardized pulp canal files of the rhomboidal design for use as cutting instruments for gaining access to, cleaning, enlarging, and smoothing the walls of a root canal during endodontic procedures.

Salient Characteristics:

Classification. Rhomboidal files shall be of the following classes and sizes:

Class 1 - 21 millimeter, operative end [1.]
Class 2 - 25 millimeter, operative end
Class 3 - 30 millimeter, operative end

Sizes - 8, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 70, 80, 90, 100, 110, 120, 130, 140

[1.] Class 1 files shall be furnished in sizes 8 through 60 only.

Material.

Operative end. The operative end of the rhomboidal files shall be fabricated from type 302 or 304 stainless steel in accordance with ASTM Designation F899 for composition.

Handle. Handles shall be fabricated of plastic material which is rigid and of a quality suitable to withstand normal operative procedures. Handles shall be capable of being autoclaved or dry heat sterilized without losing the shape of the handle or leaching out of the color when sterilized as specified in ADA Specification No. 28.

AMSC N/A FSC 6520

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Tensile strength. Rhomboidal file sizes 08 through 20 shall have a minimum tensile strength of 310,000 pounds per square inch (psi) and a minimum tensile strength of 270,000 psi for sizes 25 through 140 files. Tensile strength shall be performed in a tension tester on an 8-inch sample of steel stock used to produce the file. Average a minimum of three specimens from each lot of steel stock.

Design and dimensions. Rhomboidal files shall be of two-piece construction consisting of an operative end and a handle. The design and dimensions of the rhomboidal files shall be of the standardized system conforming to the requirements of figure 1 and table I (dimensions). The diameters (D1, D2 and D3), tip length and other dimensions shall be measured using an optical comparator, shadow-graph, or other suitable device capable of measuring with the accuracy specified. The taper shall be obtained by obtaining the diameter at 1 mm increments along the cutting section beginning 1 mm from the tip point up to 6 mm from the tip point.

Operative ends. Operative ends of instruments shall be made of stainless steel wire conforming to the requirements specified under materials. The operative end shall be uniformly twisted to the number of spirals specified in table I to form a sharp cutting instrument. Spirals forming the cutting edges shall be right hand and shall extend to the tip. The cutting edge angle (figure 1) shall be as specified in table I. The file shall have a rhomboidal cross section featuring high flutes to provide sharper cutting edges and low flutes for greater debris removal. Operative ends of class 1 instruments shall be 21 $\pm$ 1 mm in length and class 3 instruments shall have operative ends 30 $\pm$ 1 mm in length.

Handles. Operative ends of files shall be securely and permanently affixed in the handle with no movement of operative ends permitted. The entire handle shall be longitudinally serrated or shall have cross cut serrations. As an alternate, only the finger gripping portion shall be serrated. Color shall be incorporated in the plastic composition. The handle shall be in accordance with figure 1 of this description.

Taper. The taper, as measured over the larger diameter cutting edges of the rhomboidal file shall be 0.02 mm per millimeter of cutting length.

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Tip. The tip length of the instrument (pointed portion of the operative end) shall be no greater than twice the diameter of D1 of the respective size number. The tip of the instrument shall be sharply pointed.

TABLE I. Dimensions.

Diameter (Tolerance ± 0.2mm)					
Size	D1mm	D2mm	D3mm	Number of Spirals [2] ± 1	Angle of Cutting Edge ± 1 Degree
08	0.08	0.14	0.40	20	87.5
10	0.10	0.16	0.42	20	97.5
15	0.15	0.21	0.47	19	85
20	0.20	0.26	0.52	18	82.5
25	0.25	0.31	0.57	18	75
30	0.30	0.36	0.62	16	75
35	0.35	0.41	0.67	15	75
40	0.40	0.46	0.72	14	75
45	0.45	0.51	0.77	13	75
50	0.50	0.56	0.82	13	75
55	0.55	0.61	0.87	12	70
60	0.60	0.66	0.92	11	65
70	0.70	0.76	1.02	11	60
80	0.80	0.86	1.12	10	60
90	0.90	0.96	1.22	9	60
100	1.00	1.06	1.32	8	60
110	1.10	1.16	1.42	8	60
120	1.20	1.26	1.52	7	60
130	1.30	1.36	1.62	7	60
140	1.40	1.46	1.72	6	60

2. Large cutting edge only.

Resistance to fracture by twisting. The maximum torsional load before failure and the angular deflection at failure shall not be less than that given in table II for each size when tested as specified in ADA Specification No. 28.

Stiffness. The stiffness of all instruments shall not be greater than that given in table III for each size when tested as specified in ADA Specification No. 28.

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TABLE II.
Torque and Angular Deflection (Minimum Values) Twisted in Clockwise Direction.

Size	Torque (gm. cm.)	Angular Deflection (Degrees)
08	5.0	360
10	6.0	360
15	8.0	360
20	18.0	360
25	30.0	360
30	45.0	360
35	65.0	360
40	100.0	360
45	120.0	360
50	170.0	360

TABLE III.
Stiffness (Bending Moment) Maximum Values at 45^0 Bend.

Size	Bending Torque (gm. cm.)
08	16
10	20
15	35
20	65
25	80
30	100
35	170
40	200
45	350
50	400

Corrosion resistance. The operative ends of the files shall show no evidence of corrosion on their surface when subjected to the corrosion tests as specified in ADA Specification No. 28.

Finish. The spiral section of the operative end of the file shall have a bright and smooth finish. The instrument shall be uniform and free from burrs, scales, rough areas and grind marks.

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Identification marking. The handle shall be color coded and numbered with the respective size. Size and color shall be in accordance with ADA Specification No. 28.

Workmanship. The rhomboidal files shall be properly constructed and finished and shall be free from defects which may detract from their appearance or impair their intended use.

Unit. Package (PG). One package containing six pulp canal files of the class and size specified, constitutes one unit. The files shall be packaged in a plastic tray with a clear plastic slip-over or hinged cover. The tray shall be designed with six cavities or inserts to hold each file in such a manner so as to prevent movement and damage to the operative ends. The cover shall be secured with tape.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit or product for inspection purposes shall be one file.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified below. The unit of product for test purposes shall be one file.

Test Criteria

Component	Characteristic	Inspection level	AQL
	Tensile Strength	3.	---
Operative end	Dimensions	S-2	4.
	Resist. to frac.	S-2	4.
	Stiffness	S-2	4.
	Corrosion resist.	S-2	4.
Handles	Sterilization	5.	4.

3. A sufficient quantity of each lot of raw material used in the fabrication of the files shall be selected for testing.

4. Acceptance number shall be zero for all sample sizes.

5. Separate S-2 level quantities for both autoclave and dry heat tests.

Tests. Tests, as specified herein or in ADA Specification No. 28 shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification	Class	Size
6520-01-180-3567	FILE, PULP CANAL, DENTAL	1	8
6520-01-180-3568	Rhomboidal, Standardized,	1	10
6520-01-180-3569	21 mm End Length, 6s	1	15
6520-01-180-3570		1	20
6520-01-180-3571		1	25
6520-01-179-5804		1	30
6520-01-180-3573		1	35
6520-01-179-5805		1	40
6520-01-180-3572		1	45

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6520-01-179-5806	FILE, PULP CANAL, DENTAL	1	50
6520-01-179-5807	Rhomboidal, Standardized,	1	55
6520-01-179-5808	21 mm End Length, 6s	1	60
6520-01-179-5809	FILE, PULP CANAL, DENTAL,	2	8
6520-01-179-5810	Rhomboidal, Standardized,	2	10
6520-01-179-5811	25 mm End Length, 6s	2	15
6520-01-179-5812		2	20
6520-01-180-3574		2	25
6520-01-180-3575		2	30
6520-01-180-6949		2	35
6520-01-180-3243		2	40
6520-01-180-3244		2	45
6520-01-180-3245		2	50
6520-01-180-3246		2	55
6520-01-180-3247		2	60
6520-01-180-3248		2	70
6520-01-181-3187		2	80
6520-01-180-3249		2	90
6520-01-180-3250		2	100
6520-01-180-3251		2	110
6520-01-180-3252		2	120
6520-01-180-3253		2	130
6520-01-180-3254		2	I40
6520-01-176-6189	FILE, PULP CANAL, DENTAL,	3	8
6520-01-176-6190	Rhomboidal, Standardized,	3	10
6520-01-176-6191	30 mm End Length, 6s	3	15
6520-01-176-6192		3	20
6520-01-176-6193		3	25
6520-01-176-6194		3	30
6520-01-176-7531		3	35
6520-01-176-7530		3	40
6520-01-176-7532		3	45
6520-01-176-7533		3	50
6520-01-176-7534		3	55
6520-01-176-4633		3	60
6520-01-176-4632		3	70
6520-01-176-4634		3	80
6520-01-176-4635		3	90
6520-01-176-4636		3	100
6520-01-182-3923		3	110
6520-01-177-0131		3	120
6520-01-177-0132		3	130
6520-01-176-7611		3	140

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MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

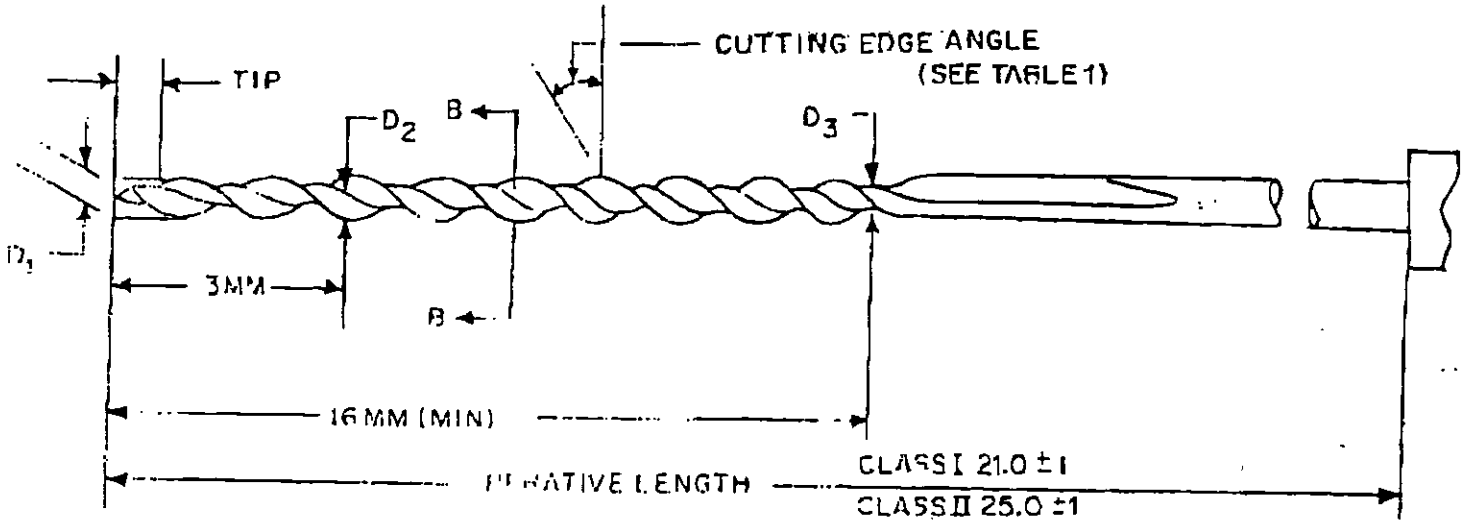
DLA-DM

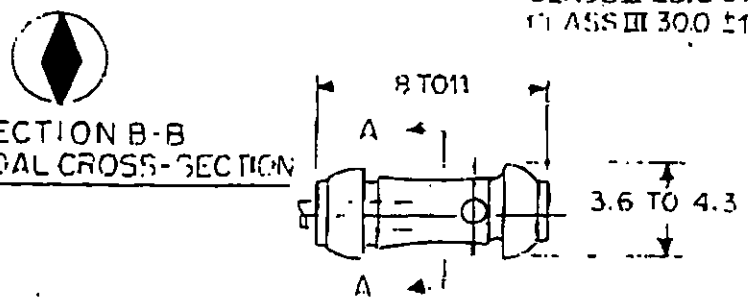
CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

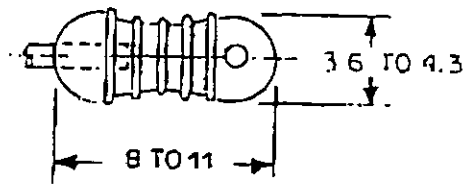
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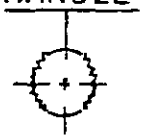


SECTION B-B
RHOMBOIDAL CROSS-SECTION

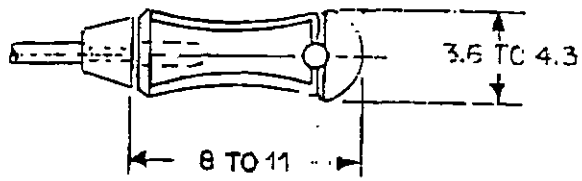


ALTERNATE HANDLE

HANDLE



SECTION A-A



ALTERNATE HANDLE

NOTES:

1. HANDLES SHALL BE IN ACCORDANCE WITH THIS FIGURE FOR DIMENSIONAL REQUIREMENTS; SHAPE OF HANDLES SHOWN ARE TYPICAL AND ARE NOT INTENDED TO RESTRICT OR PRECLUDE THE PURCHASE OF FILES WHICH ARE OTHERWISE IN ACCORDANCE WITH THIS SPECIFICATION.
2. HOLE IN HANDLE SHALL BE OPTIONAL.
3. ALL DIMENSIONS ARE IN MILLIMETERS

FIGURE 1 - FILE PULP CAN/ RHOMBOIDAL

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